Feature Engineering for Professional Tennis Match Prediction

This project aims to develop a machine learning framework to predict the outcome of professional tennis matches by leveraging a comprehensive historical dataset. While standard features such as ATP (Association of Tennis Professionals) rankings, tournament tier, and court surface provide a baseline, they often fail to capture the nuances of player "form" (recent results) and specific tactical advantages. To address this, this project will focus on engineering "enhanced features". One possible idea is to use an adaptive Elo rating system. Unlike traditional rankings that reward deep tournament runs, the Elo system (originally designed for chess) quantifies a player's skill relative to the strength of their opponents. By implementing surface-specific Elo ratings (e.g., separate Elo rankings for clay, grass, and hard courts) and giving more importance to most recent results, the model should more accurately reflect a player's current threat level on a specific court type.

Beyond Elo ratings, the project will explore the predictive power of tactical and physiological "interaction features". This includes modelling match-up dynamics, such as the historical performance of left-handed players against right-handed opponents, and quantifying player-specific surface sensitivities and rivalries. The methodology will involve training several classifiers, including Random Forest and XGBoost, and deep learning techniques. The goal is to determine if engineered features can improve baseline models and bridge the gap between statistical predictions and professional betting market odds, providing deeper insights into the variables that truly dictate success on the professional tour.

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